Hao Zhu

PhD student

Neurorobotics Group
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https://haozhu10015.github.io

— Education

2024– **PhD in Computer Science**, University of Freiburg, Germany
Advisors: Prof. Joschka Boedecker, Prof. Moritz Diehl

2021–2023 **M.Sc. in Neuroscience**, University of Freiburg, Germany
GPA: 3.8 | German grade: 1.3
Thesis: *Deciphering Decision Making with Inverse Reinforcement Learning*
Advisors: Prof. Ilka Diester, Prof. Joschka Boedecker

2016–2020 **B.Sc. Chemical Biology**, Nankai University, China
GPA: 3.6 | Average grade: 89.4/100
Thesis: *Identification of Functional Residues in the Human Protoporphyrinogen Oxidase with the Network Model and Site-Directed Mutagenesis*
Advisors: Prof. Xin Wen, Prof. Zhen Xi

— Academic experience

2024– **Researcher**, Neurorobotics Group, University of Freiburg, Germany
Part of the project IN-CODE funded by the German Research Foundation (DFG) for developing optimization theory, methods, and software for scientific machine learning.

2022–2023 **Research intern**, Optophysiology Lab, University of Freiburg, Germany
Led a research project on the mathematical modeling of rodent decision making behavior via (inverse) reinforcement learning.

2022 **Research intern**, Straw Lab, University of Freiburg, Germany
Led a research project on developing Kalman filter auto-tuning algorithm for animal tracking, and designing event-camera-based lock-on tracker prototype, steering multiple cameras for tracking bees in the wild.

2017–2021 **Undergraduate researcher/research assistant**, State Key Laboratory of Elemento-Organic Chemistry, Nankai University, China
Led the National Training Program of Innovation and Entrepreneurship for Undergraduates project titled *Computational Simulation and Biological*

Verification for Different Species of Protoporphyrinogen IX Oxidase Amino Acid Interactions, aiming at identifying key amino acid residues in protoporphyrinogen oxidase with computational quantum mechanics, molecular dynamics simulation, and graph theory.

— Teaching

Lecture

Materials of all my classes are made public at `https://haozhu10015.github.io/teaching`.

2026– **Head lecturer**, *Disciplined Machine Learning*, University of Freiburg

This is a new course that is now being developed by me and Prof. Joschka Boedecker, and is to be taught in 2026.

2024– **Co-lecturer**, *Probabilistic Graphical Models*, University of Freiburg

I developed the materials for this course in 2024 and have co-taught it with Prof. Joschka Boedecker every summer since then.

Seminar

2026 **Supervisor**, *Causal World Models*, University of Freiburg

2024 **Supervisor**, *Causality and Reinforcement Learning*, University of Freiburg

— Software

Below is a selection of my software projects. A complete list of projects I have led or contributed to is available at `https://haozhu10015.github.io/software`.

- `dbcp`: a CVXPY extension for disciplined biconvex programming
- `cvxrust`: a Rust implementation of the disciplined convex programming framework; under development
- `jemdoc-rs`: a Rust rewrite of the static site generator `jemdoc` for creating academic personal websites

— Awards

2020 Innovative Scientific Research Award for College Students, Nankai University (Excellence Award)

2018 Asymchem Scholarship of Chemistry College, Nankai University

2017 Asymchem Scholarship of Chemistry College, Nankai University

— Invited talks

2026 **CVXPY Workshop**, Stanford University
Title: *Disciplined biconvex programming*

2024 **Bernstein Center Freiburg Seminar**, University of Freiburg
Title: *Inverse reinforcement learning in animal behavior characterization*

— Publications

The following publications are up to date as of March 14, 2026. A more frequently updated list of my publications is available at `https://haozhu10015.github.io/papers`.

Book

- H. Zhu and J. Boedecker. *Disciplined Machine Learning*, 2026. Working draft, available at `https://haozhu10015.github.io/books/dmlbook`.

Papers as first author

- H. Zhu and J. Boedecker. Disciplined biconvex programming. *arXiv*, 2511.01813, 2025. Under review, *Journal of Optimization Theory and Applications*.
- H. Zhu, J. Hoffmann, B. Zhang, and J. Boedecker. Fitting reinforcement learning model to behavioral data under bandits. *arXiv*, 2511.04454, 2025. To appear, *Frontiers in Applied Mathematics and Statistics*.
- H. Zhu, S. Yan, J. Hoffmann, and J. Boedecker. Multi-convex programming for discrete latent factor models prototyping. *arXiv*, 2504.01431, 2025. Under review, *Optimization and Engineering*.
- H. Zhu and J. Boedecker. Solving inverse problem for multi-armed bandits via convex optimization. *arXiv*, 2501.18945, 2025.
- H. Zhu, Y. Zhang, and J. Boedecker. Inverse reinforcement learning via convex optimization. *arXiv*, 2501.15957, 2025.
- H. Zhu, B. De La Crompe, G. Kalweit, A. Schneider, M. Kalweit, I. Diester, and J. Boedecker. Multi-intention inverse Q-learning for interpretable behavior representation. *Transactions on Machine Learning Research*, 2024.

Papers as co-author

- B. Zhang, Y. Zhang, H. Zhu, S. Yan, T. Brox, and J. Boedecker. Constrained reinforcement learning with smoothed log barrier function. *Transactions on Machine Learning Research*, 2025.
- K. Kleespies, P. C. Paulus, H. Zhu, F. Pargent, M. Jakob, J. Werle, M. Czisch, J. Boedecker, S. Gais, and M. Schönauer. Sleep resolves competition between explicit and implicit memory systems. *bioRxiv*, 2025.

- B. Wang, Z. Zhang, H. Zhu, C. Niu, X. Wen, and Z. Xi. The hydrogen bonding network involved Arg59 in human protoporphyrinogen IX oxidase is essential for enzyme activity. *Biochemical and Biophysical Research Communications*, 557:20–25, 2021.

— References

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